

ABHISHEK GOYAL

[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / %
Ongoing	M.Sc in Mathematics	Indian Institute of Technology Delhi	7.8
2025	B.Sc-B.Ed (Integrated)	University of Rajasthan	73%
2021	CBSE	SR International Academy, Nadbai	88.2%

SCHOLASTIC ACHIEVEMENTS

- Joint Admission Test (JAM) Rank:** Secured **AIR 202** in Mathematics
- Department Rank:** Ranked **16th** in the Department of Mathematics, IIT Delhi

TECHNICAL SKILLS

- Languages:** Python, SQL, C, HTML/CSS
- ML/NLP:** Scikit-learn, XGBoost, CatBoost, Optuna, NLTK, SpaCy, TextBlob
- Deep Learning:** PyTorch, TensorFlow, Keras, Transformers, Hugging Face
- Data / Visualization:** Pandas, Numpy, Matplotlib, Seaborn
- MLOps & Deployment:** Docker, GitHub Actions, Streamlit, Gradio, Hugging Face Spaces, GitHub Pages

PROJECTS

Signature Forgery Verification: Deep Metric Learning for Writer-Independent Verification [Demo](#) | [GitHub](#)

- Data Leak Mitigation:** Exposed two systemic data leaks in the public ICDAR benchmark (byte-identical duplicates and deterministic folder labels), correcting naïve ROC-AUC from 0.999 to an honest 0.973 via an automated script.
- Model Progression:** Evolved through 4 metric-learning paradigms (contrastive → triplet → fine-tuned EfficientNet-B0 → **online batch-hard mining**), achieving **Test ROC-AUC 0.986**, EER 8.0%, FAR 6.5%, and a **cross-dataset AUC of 0.877** on unseen writers.
- Adaptive Evaluation:** Implemented per-writer adaptive thresholds ($\tau_w = \mu + \alpha \cdot \sigma$, tuned on validation) reaching a FAR of **3.5%** and FRR of 7.7%; successfully extended evaluation to cross-script (Latin + Devanagari) datasets.

CCPA Dark Pattern Detector: NLP Baseline vs. Fine-Tuned Transformer [Classical](#) | [Transformer](#) | [GitHub](#)

- Compliance Auditor:** Mapped UI text into **14 CCPA classes** across a 6.7K-sample corpus; identified and resolved a near-duplicate train/test leak using **skeleton-aware splitting**, correcting macro-F1 from ~0.94 to an honest ~0.64.
- Feature Engineering:** Extracted **22 NLP features** (TextBlob sentiment, keyword triggers, structural proxies) utilizing TF-IDF, Robust Scaling, and Yeo-Johnson transformations; optimized XGBoost using **Optuna** with SMOTE in a Stratified 5-Fold CV.
- Transformer Fine-Tuning:** Fine-tuned **DistilBERT**, achieving a **macro-F1 of 0.797** (OOD: 0.594 on real Indian UIs) compared to a Classical SVC baseline (0.654 / OOD: 0.475); deployed as both **Streamlit** and **HF Spaces** apps.

Indian Tech Salary Predictor: MLOps-Instrumented Compensation Analytics Engine [Demo](#) | [GitHub](#)

- Data Pipeline:** Cleaned **110K records** of Indian tech salary data by standardizing synonyms, parsing multi-label skills into binary flags, and imputing outliers within a reproducible scikit-learn **ColumnTransformer** pipeline.
- Model & MLOps:** Benchmarked 4 models (ElasticNet, Random Forest, XGBoost, CatBoost), deploying a **Tuned CatBoost** ($R^2 = 0.8753$, MAE $\approx$ INR 1.60 LPA) for a 34% baseline improvement; instrumented the lifecycle with **MLflow**, **DVC**, and **Docker**.

CERTIFICATIONS

Machine Learning Specialization

May 2026

Stanford University / DeepLearning.AI (Coursera)

Supervised learning, Unsupervised learning, Recommender systems, Artificial neural networks

Deep Learning Specialization

Jun 2026

DeepLearning.AI (Coursera)

Neural Networks, CNNs, RNNs, Seq2Seq, Attention Models, Transformers

IBM RAG & Agentic AI Professional Certificate

Jul 2026

IBM (Coursera)

Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Agentic AI, Vector Databases